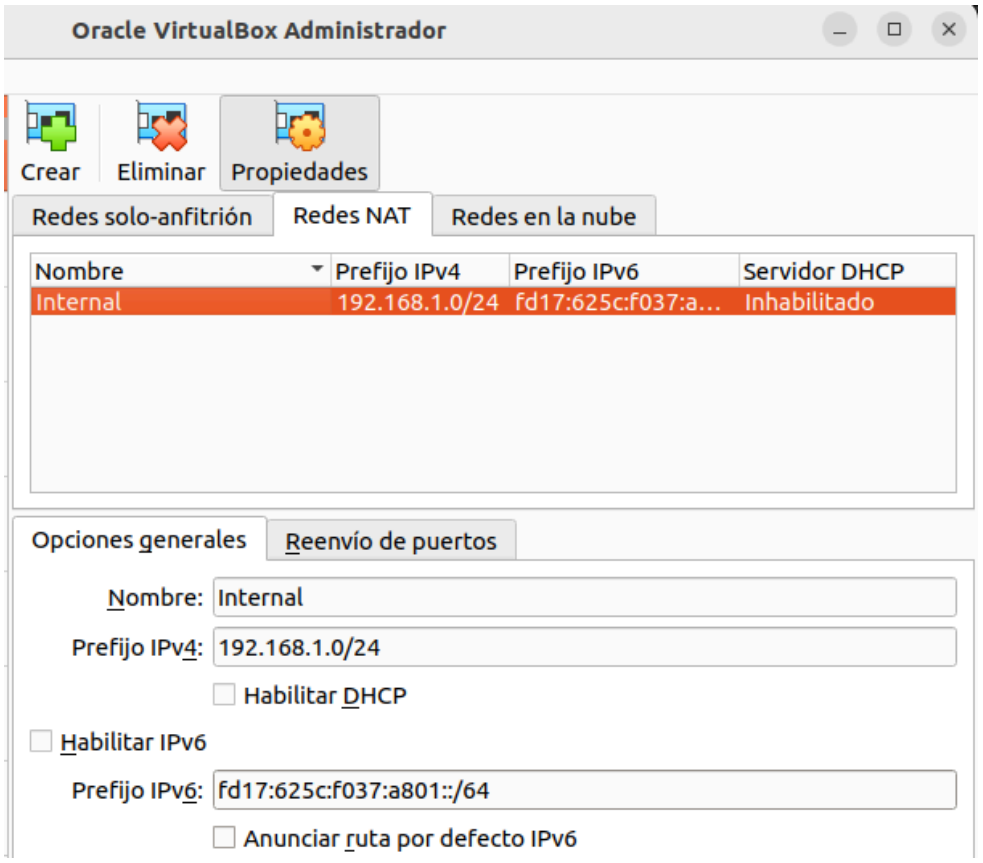
1. Preparar máquina virtual Ubuntu Server con al menos una interfaz de red en modo interno

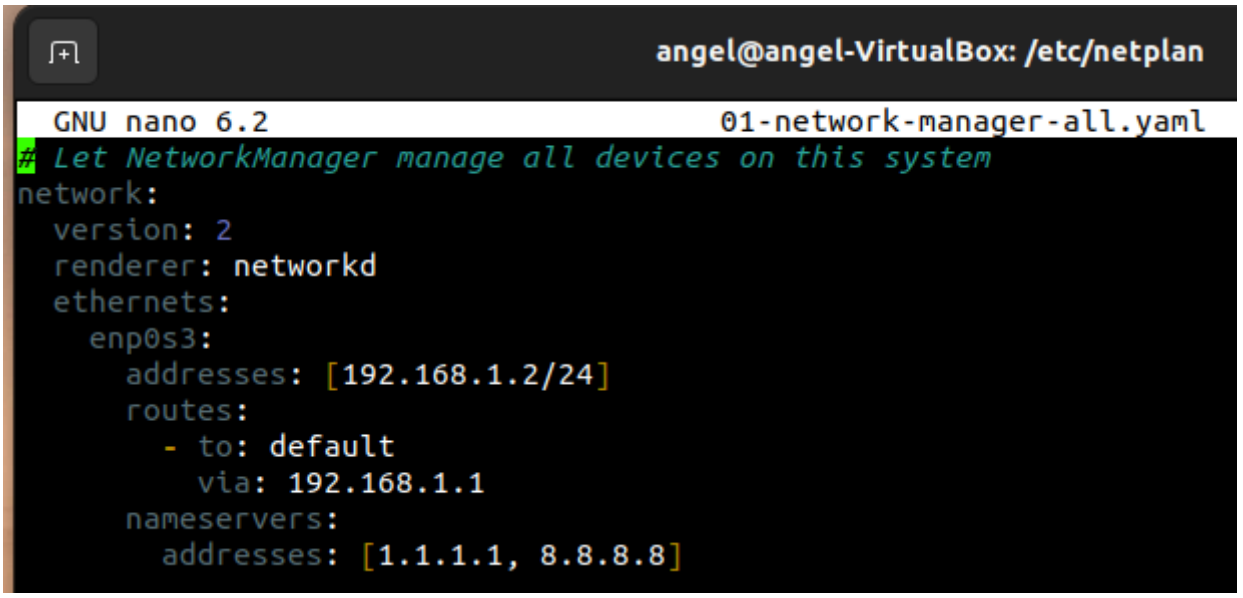


2. Instalar servidor DHCP isc-dhcp-server

sudo apt update

sudo apt install isc-dhcp-server -y

Se configura el archivo **01-network-manager-all.yaml** dentro del directorio **/etc/netplan/** de la siguiente manera:



3. Configurar servidor DHCP a través del fichero dhcpd.conf para servir un rango libre (ej. de 192.168.1.20 a 192.168.1.40 con máscara /24)

4. Configurar los parámetros de los clientes con al menos las siguientes opciones:
1. Router por defecto: 192.168.1.1
2. Nombre de dominio: smr.local
3. Servidores DNS: 172.20.30.1, 172.20.30.2 u 8.8.8.8, 1.1.1.1
4. Tiempo de concesión por defecto: 15 minutos
5. Tiempo de concesión máximo: 2 horas

Editamos **/etc/default/isc-dhcp-server** de la siguiente manera:

```
angel@angel-VirtualBox: /etc/netplan
  GNU nano 6.2                    /etc/default/isc-dhcp-server
# Defaults for isc-dhcp-server (sourced by /etc/init.d/isc-dhcp-server)

# Path to dhcpd's config file (default: /etc/dhcp/dhcpd.conf).
#DHCPDv4_CONF=/etc/dhcp/dhcpd.conf
#DHCPDv6_CONF=/etc/dhcp/dhcpd6.conf

# Path to dhcpd's PID file (default: /var/run/dhcpd.pid).
#DHCPDv4_PID=/var/run/dhcpd.pid
#DHCPDv6_PID=/var/run/dhcpd6.pid

# Additional options to start dhcpd with.
#       Don't use options -cf or -pf here; use DHCPD_CONF/ DHCPD_PID instead
#OPTIONS=""

# On what interfaces should the DHCP server (dhcpd) serve DHCP requests?
#       Separate multiple interfaces with spaces, e.g. "eth0 eth1".
INTERFACESv4="enp0s3"
INTERFACESv6=""
```

```
angel@angel-VirtualBox: /etc/netplan
  GNU nano 6.2                          /etc/dhcp/dhcpd.conf
# dhcpd.conf
#
# Sample configuration file for ISC dhcpd
#
# Attention: If /etc/ltsp/dhcpd.conf exists, that will be used as
# configuration file instead of this file.
#

# option definitions common to all supported networks...
option domain-name "smr.local";
option domain-name-servers ns1.example.org, ns2.example.org;

default-lease-time 900;
max-lease-time 7200;
```

```
angel@angel-VirtualBox: /etc/netplan
  GNU nano 6.2                          /etc/dhcp/dhcpd.conf
# No service will be given on this subnet, but declaring it helps the
# DHCP server to understand the network topology.

#subnet 10.152.187.0 netmask 255.255.255.0 {
#}

# This is a very basic subnet declaration.

subnet 192.168.1.0 netmask 255.255.255.0 {
  range 192.168.1.20 192.168.1.40;
  option routers 192.168.1.1;
  option subnet-mask 255.255.255.0;
  option domain-name-servers 172.20.30.1 , 172.20.30.2;
}
```

5. Configurar también al menos una reserva estática para un equipo (ej. 192.168.1.5)

```
angel@angel-VirtualBox: ~
  GNU nano 6.2                          /etc/dhcp/dhcpd.conf *
# allocated dynamically (if possible), but the host-specific info
# will still come from the host declaration.

host ucliente {
  hardware ethernet 08:00:27:9c:e2:90;
  fixed-address 192.168.1.5;
#  filename "vmunix.passacaglia";
#  server-name "toccata.example.com";
}
```

para activar el servicio networkd
sudo systemctl restart systemd-networkd
para cambiar la configuración de red
sudo netplan apply

Ahora comprobamos que el servidor DHCP está activo

```
angel@angel-VirtualBox: /etc/netplan

Use «fg» para volver a nano.

[1]+  Detenido                 sudo nano /etc/dhcp/dhcpd.conf
angel@angel-VirtualBox:/etc/netplan$ sudo nano /etc/dhcp/dhcpd.conf
angel@angel-VirtualBox:/etc/netplan$ sudo nano /etc/dhcp/dhcpd.conf
angel@angel-VirtualBox:/etc/netplan$ sudo service isc-dhcp-server restart
angel@angel-VirtualBox:/etc/netplan$ sudo service isc-dhcp-server status
● isc-dhcp-server.service - ISC DHCP IPv4 server
     Loaded: loaded (/lib/systemd/system/isc-dhcp-server.service; enabled; vendor preset: ena
     Active: active (running) since Tue 2024-11-19 19:05:07 CET; 10s ago
       Docs: man:dhcpd(8)
   Main PID: 5754 (dhcpd)
      Tasks: 4 (limit: 6679)
     Memory: 4.5M
        CPU: 8ms
     CGroup: /system.slice/isc-dhcp-server.service
             └─5754 dhcpd -user dhcpd -group dhcpd -f -4 -pf /run/dhcp-server/dhcpd.pid -cf /

nov 19 19:05:07 angel-VirtualBox dhcpd[5754]: PID file: /run/dhcp-server/dhcpd.pid
nov 19 19:05:07 angel-VirtualBox dhcpd[5754]: Wrote 0 leases to leases file.
nov 19 19:05:07 angel-VirtualBox sh[5754]: Wrote 0 leases to leases file.
nov 19 19:05:07 angel-VirtualBox dhcpd[5754]: Listening on LPF/enp0s3/08:00:27:24:35:9b/192.1
nov 19 19:05:07 angel-VirtualBox sh[5754]: Listening on LPF/enp0s3/08:00:27:24:35:9b/192.168.
nov 19 19:05:07 angel-VirtualBox sh[5754]: Sending on   LPF/enp0s3/08:00:27:24:35:9b/192.168.
nov 19 19:05:07 angel-VirtualBox sh[5754]: Sending on   Socket/fallback/fallback-net
nov 19 19:05:07 angel-VirtualBox dhcpd[5754]: Sending on   LPF/enp0s3/08:00:27:24:35:9b/192.1
nov 19 19:05:07 angel-VirtualBox dhcpd[5754]: Sending on   Socket/fallback/fallback-net
nov 19 19:05:07 angel-VirtualBox dhcpd[5754]: Server starting service.
lines 1-21/21 (END)
```

6. Probar la funcionalidad con el cliente Ubuntu

```
angel@angel-VirtualBox: /etc/netplan
    link/ether 08:00:27:9c:e2:90 brd ff:ff:ff:ff:ff:ff
angel@angel-VirtualBox:/etc/netplan$ sudo systemctl restart systemd-networkd
angel@angel-VirtualBox:/etc/netplan$ sudo netplan apply

** (generate:2317): WARNING **: 20:30:28.048: Permissions for /etc/netplan/01-network-manager-all.ya
ml are too open. Netplan configuration should NOT be accessible by others.

** (process:2315): WARNING **: 20:30:28.271: Permissions for /etc/netplan/01-network-manager-all.yam
l are too open. Netplan configuration should NOT be accessible by others.

** (process:2315): WARNING **: 20:30:28.372: Permissions for /etc/netplan/01-network-manager-all.yam
l are too open. Netplan configuration should NOT be accessible by others.

** (process:2315): WARNING **: 20:30:28.372: Permissions for /etc/netplan/01-network-manager-all.yam
l are too open. Netplan configuration should NOT be accessible by others.
angel@angel-VirtualBox:/etc/netplan$ ip ad
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 100
0
    link/ether 08:00:27:9c:e2:90 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.20/24 brd 192.168.1.255 scope global dynamic noprefixroute enp0s3
       valid_lft 896sec preferred_lft 896sec
    inet6 fe80::a00:27ff:fe9c:e290/64 scope link
       valid_lft forever preferred_lft forever
angel@angel-VirtualBox:/etc/netplan$
```

7. Verificar que los parámetros de configuración en los clientes son los correctos

```
angel@angel-VirtualBox: /etc/netplan
  GNU nano 6.2                          01-network-manager-all.yaml
# Let NetworkManager manage all devices on this system
network:
  version: 2
  renderer: NetworkManager
  ethernets:
    enp0s3:
      dhcp4: true
```